Hartshorne 3-02-2025

Chapter II.4-5, Week 6

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4.9. Show that a composition of projective morphisms is projective. Conclude that projective morphisms have all properties listed in Ex. 4.8.

Proof. □

5.1. f

Proof. □

5.3. Let $X = \operatorname{Spec} A$ be an affine scheme. Show that the functors $\widetilde{(-)}$ and Γ are adjoint. That is, for any A-module M, and for any sheaf of $\mathcal{O}_X$-modules $\mathscr{F}$, there is a natural isomorphism

$$\operatorname{Hom}_A(M, \Gamma(X, \mathscr{F})) \cong \operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathscr{F})$$

Proof. □

5.16. For $(X, \mathcal{O}_X)$ a ringed space and $\mathscr{F}$ a sehaf of $\mathcal{O}_X$-modules, we define the *tensor algebra*, *symmetric algebra*, and *exterior algebra* of $\mathscr{F}$ by taking the sheaves associated to the associated presheaf.

(a) Suppose that $\mathscr{F}$ is locally free of rank n. Then $T^r(\mathscr{F})$, $S^r(\mathscr{F})$ and $\wedge^r(\mathscr{F})$ are locally free, of ranks rn, $\binom{n+r-1}{n-1}$, and $\binom{n}{r}$ respectively.

(b) Again, let $\mathscr{F}$ be locally free of rank n. Then the multiplication map

5.17.

5.18.